

Active View Planning for Counting Apples in Orchards

Pravakar Roy and Volkan Isler

Abstract—We consider an agricultural automation scenario where a robot, equipped with a camera mounted on a manipulator, is charged with counting the number of apples in an orchard. We focus on the subtask of planning views so as to accurately estimate the number of apples in an apple cluster. We present a method to efficiently enumerate combinatorially distinct world models and to compute the most likely model from one or more views. These are incorporated into single and multi-step planners. We evaluate these planners in simulation as well as with experiments on a real robot.

I. INTRODUCTION

Yield mapping is emerging as an important task in agricultural automation [1]–[4]. Current systems for automated yield mapping involve the capture of imagery along an arbitrary and/or predetermined path. This approach is sufficient for large commercial orchards which have “fruit-walls” where the fruits are visible from the outside and the clusters contain a small number of fruits. However, in more general settings, estimating the exact count from an arbitrary view is difficult. See Figure 1.

In this work, we show how active vision techniques can be used to accurately obtain the number of apples in a cluster. Consider a system where a robot equipped with a controllable camera is charged with surveying an orchard (Fig. 1 (rightmost image)). Imagine that the robot positions the base near a tree and obtains the first view of an apple cluster. Where should it take the second image from so as to accurately estimate the number of apples in the cluster? We study this novel active perception problem.

Our approach can be summarized as follows: Consider the set of all possible “world states” where each world state encodes the position and diameter of each apple. The images acquired by the robot can be used to associate likelihoods to these world views. The goal is to plan views that reduce the entropy of this distribution sufficiently so that we can estimate the count accurately. Directly enumerating all such world states is computationally expensive: In a $1m \times 1m \times 1m$ volume, a $1cm$ resolution grid would have 10^6 cells. Even if we assume that the diameter is fixed, for a cluster of 6 apples, there would be 10^{36} hypotheses. Our main contributions in this work are (1) a technique to efficiently enumerate only combinatorially distinct world states, and (2) a method to estimate their likelihoods from images. We experimentally demonstrate that together these methods yield an efficient view planning method for counting spherical objects which can be used in other agricultural applications.

The authors are with the Department of Computer Science and Engineering, University of Minnesota, Minneapolis, MN, USA. {royxx268, isler}@umn.edu.

The paper is organized as follows: We present an overview of related work in section II. In section III, we present our system description, which is followed by the problem formulation in section IV. From section V to section VI, we present our technical approach which includes generating the world models and planning for the next best view. We present our simulation and indoor experiment results in section VII followed by the conclusion and future work.



Fig. 1: A single view of a cluster can be deceiving (First three images from left). The first view suggests that there are three apples, the second one suggests five and finally from a bottom up view we see that there are actually six apples in the cluster. The rightmost image shows our platform for automating different precision agriculture and phenomics operation.

II. RELATED WORK

In active vision, the parameters of a camera are actively controlled so as to improve the performance of the vision system [5]. One of the fundamental problems in active vision is view planning to improve coverage [6]. Recently, there has been increasing activity to scale up active perception to sensing tasks over longer periods of time and in larger environments [7]. Recent work in this area is concerned with object detection and classification [8]–[10]. In the context of agricultural applications, most of the work on yield mapping does not involve active perception [1]–[4]. Related to our work, Baur et al. [11] designed a heuristic based algorithm for trajectory generation for apple picking. Underwood et al. [12] developed a system for weed removal. In this work, we explicitly address the occlusions among objects (the apples) and present representations and planning algorithms to estimate their count.

III. SYSTEM DESCRIPTION

Our system consists of a monocular camera mounted on a robotic manipulator which is fixed on a ground robot (Fig. 1 (rightmost image)). Our long-term target is to develop algorithms enabling this platform to traverse the orchard rows autonomously and collect useful data for phenomics

and precision agriculture. This paper focuses specifically on close up view planning for fruit counting using the camera mounted on the manipulator. We assume that the camera is calibrated. The robot pose ${}^wR_r[t_r]$ and the camera pose ${}^wR_c[t_c]$ are known with respect to world frame $\{w\}$. From this camera $\{c\}$ mounted on a robotic manipulator $\{r\}$ and facing an apple cluster A , we want to compute a sequence of viewpoints to place the camera to correctly predict the number of apples. We assume that we have a rough estimate (d_{est}) of the working distance (distance from the camera to the apple cluster). Using this estimate, we define a viewing sphere S centered at the center of A and having radius d_{est} . As the viewing sphere is around the cluster center, we can only operate in the hemisphere facing the camera which is in the reachable workspace of the robot. We place m viewpoints $V = \{v_1, v_2, \dots, v_m\}$ in the frontal hemisphere of S . Fig. 2 shows an example of one such viewing sphere (left image) and the viewpoints placed on it (right image). The ellipsoids inside the viewing sphere are colored randomly for clarity of understanding and have no other significance.

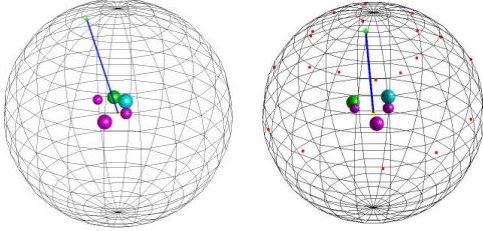


Fig. 2: We define the center of our viewing sphere to be the perceived center of the apple cluster under observation. The radius is set as the estimated distance from camera to the cluster center. Figure on left shows the viewing sphere, the ellipsoidal fruits inside it (the fruits inside the cluster are colored randomly for identification purposes) and the blue line joining camera center to the cluster center. Figure on the right, shows the placed viewpoints (red dots).

IV. PROBLEM FORMULATION

In this section, we formalize the problem definition for intra-cluster view planning. The term, “intra-cluster” means that at any specific time instant we are only planning views for one particular cluster. We will represent the apples in 3D space by their enclosing ellipsoids e . Let $A = \{e_1, e_2, \dots, e_n\}$ represent an apple cluster of size n in world space. The condition for apples being in a cluster is that the maximum distance between any two enclosing ellipsoids e_i, e_j where $i, j \in [1, n]$ has to be bounded. As mentioned before, we assume that we have an approximate estimate of the working distance d_{est} from the camera to the cluster A . Given this, we define our view planning problem as the following

Problem Statement: Given an initial image I_1 and an approximate distance from camera to the cluster A , compute the next best view v_k , so as to minimize the entropy of our set of world hypotheses W ,

$$\arg \min_{v_k} H(W|I_{v_1}, I_{v_k}) = \arg \min_{v_k} - \frac{\mathcal{P}(W|I_{v_1}, I_{v_k})}{\log \mathcal{P}(W|I_{v_1}, I_{v_k})} \quad (1)$$

where $v_k \in V$ and it is not the initial view point.

V. GENERATING WORLD MODELS

In the previous section, we formulated the problem as computing a viewpoint that minimizes the entropy $H(W|I_{v_1}, I_{v_k})$. The first and most important step for our view planning strategy is generating a complete set of world hypotheses W . They must cover all possible geometric configurations for a maximum cluster size to ensure convergence to the correct fruit count.

Instead of enumerating all possible ellipsoids (which can be computationally prohibitive), we utilize a small set of combinatorial models in terms of visible and occluded apples which does not have any position and orientation associated with it. This combinatorial representation of fruits is our main contribution in this paper. We convert these models to physical world models using image based measurements. In this section, we present the details of the world model generation procedure. We start by introducing the combinatorial world models.

A. Combinatorial World Models

Two models are combinatorially equivalent if the number of visible and occluded apples from the orthographic frontal view are equal. Figure 3(b) shows two such models. We assume that the initial view obtained by the camera is the view from the frontal plane. According to our assumption $W = \{w_1, w_2, \dots, w_p\}$, is a set of all possible world models and where $w_i = \{e_1^{w_i}, e_2^{w_i}, e_3^{w_i} \dots e_n^{w_i}\}$, is a set of n ellipsoids in world space where $i \in [1, p]$ and $n \in [1, \maxClusterSize]$.

We fix the cluster size n and generate our world models in terms of occluded and visible apples. Initially, we assume that each visible apple can occlude only one other apple. The occluded apples cannot further occlude other apples. Later, we show how to extend this approach to the case where occluded apples are allowed to occlude other apples. Under the current assumption, there are only two layers: visible and occluded. The maximum number of occluded apples a_{oc} and minimum number of visible apples a_{vis} , for cluster size n and number of layers $L = 2$ is

$$a_{oc} = \left\lceil \frac{n}{L} \right\rceil$$

$$a_{vis} = n - \left\lceil \frac{n}{L} \right\rceil$$

By varying the number of visible apples (i.e the apples in layer 1) from 1 to n we can generate all possible models for this particular cluster size n . The total number of models for cluster size n is actually, $\binom{L+a_{oc}-1}{a_{oc}} = a_{oc}$. In practice, this assumption of one occlusion layer (i.e $L = 2$) holds for most apple clusters.

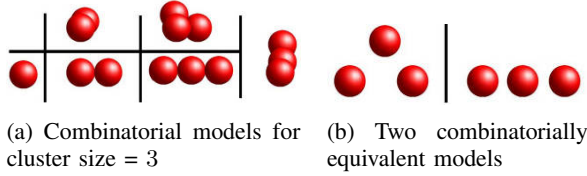


Fig. 3: Combinatorial Models. Fig.(a) shows the combinatorial models for clusters of maximum size three. It shows that there is six model in total with no restriction on the number of occlusion layers. If we restrict the number of occlusion level to one, the last model is eliminated. Right image shows two combinatorially equivalent views

We can easily extend the current method for the cases of multiple occlusion layers. Again, we assume that each apple can occlude just one apple. When the number of occluded apples is 1, or $n - 1$ where n is the cluster size, the number of models is trivially one. From the previous paragraph, we know that when the number of layers $L = 2$, the total number of models is $\binom{1+a_{oc}}{a_{oc}}$. For cases where $L > 2$ and $1 < a_{oc} < n - 1$ we find the models by using a recursive divide and conquer strategy. We subdivide occluded apples as visible and hidden for $L \geq 2$, decrease the occlusion level by 1 until we reach one of the base cases and combine the solutions together.

It should be noted that these models are not concerned with the actual locations or sizes of the apples. Their purpose is to capture the combinatorial structure of the clusters in terms of visible and occluded apples and occlusion hierarchy. Consequently, the models only need a tag for visible/occluded and a layer number. For the rest of the paper, we refer to these models as combinatorial world models. Figure 3(a) shows combinatorial world models for maximum cluster size three.

B. Sensing from Individual Images

For predicting next best views and to relate the combinatorial models to actual images we need to have a sensing method for individual images and actual physical models corresponding to each of the combinatorial model. For sensing from singular input images, we utilize our previous work [3]. There, we exploited the idea of the images being generated from a two-dimensional world model (probability distribution). Each apple in the input image is modelled by a Gaussian probability distribution function (pdf) and apple clusters are modelled as mixture of Gaussian.

For a given image and a particular size of input clusters, this method provides us with the tentative pixel location and pixel size for each apple and a likelihood of that cluster size being optimal. Essentially, our sensing method provides us with the likelihood of a certain number of apples being visible in the input image. In the next section, we look at how we can combine this information with the combinatorial world models and associate position and orientation to them.

C. From Combinatorial Models to World Models with Position and Orientation

Each of our combinatorial world models contains a nonzero number of visible apples and a few occluded apples. The sensing model provides us with an estimate of the visible apples in the image. We backproject this apples and utilize estimated depth d_{est} to obtain their tentative 3D location. The sensing estimate provides only diameters across x and y axis. We assume the diameter of the ellipsoid along the z axis is average of the x, y axis diameter. Following this procedure for all the combinatorial models, we obtain a physical model corresponding to each of the visible apples in the combinatorial models. Now, we need to associate position and orientation to the occluded apples. We place the occluded apples exactly behind the visible apples, following a left to right order. For the case of multiple level occlusion, an apple can occupy a place in a higher occlusion level if all the corresponding places in the lower occlusion levels are filled. The sensing model only provide us with likelihoods of the visible apples. To generate the likelihoods for models with occlusion we utilize the penalty and reward function described in [3]. The occluded apples do not contribute anything to the reward function and consequently they are just penalized for the increasing number of components. Therefore, by adding the penalty for the occluded apples to the likelihood of the visible apples we obtain the likelihood of the world models with occlusion. Figure 4 shows an example of the generated world models for three apples.

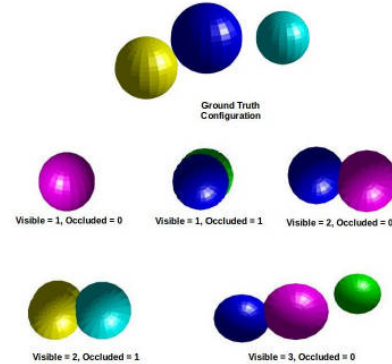


Fig. 4: World hypotheses generated from combinatorial models for three apples.

VI. VIEW PLANNING

In this section, we present the details of our view planning approach. We show how to determine the next best view point from the current information, how to update the posterior probabilities of the world models after each additional view and the termination criteria to be checked on each iteration.

At each viewpoint, the camera provides us an image of the apple cluster, $I_{v_j}(A)$, where v_j is the j^{th} viewpoint. Let, $v_1, v_2, \dots, v_k$ be the viewpoints we have chosen so far. Therefore, the next best view should be chosen so that it

minimizes the entropy of the entire set of world hypotheses W . Equation (1) has the following general form

$$\arg \min_{v_{k+1}} \sum_{i=1}^p -\mathcal{P}(w_i | I_{v_1}, I_{v_2}, \dots, I_{v_k} I_{v_{k+1}}) \quad (2)$$

$$\log \mathcal{P}(w_i | I_{v_1}, I_{v_2}, \dots, I_{v_k} I_{v_{k+1}})$$

Now we take a closer look at the term $\mathcal{P}(w_i | I_{v_1}, I_{v_2}, \dots, I_{v_k})$ and see how we can evaluate it.

$$\mathcal{P}(w_i | I_{v_1}, I_{v_2}, \dots, I_{v_k}) = \frac{\mathcal{P}(I_{v_1}, I_{v_2}, \dots, I_{v_k} | w_i) \mathcal{P}(w_i)}{\mathcal{P}(I_{v_1}, I_{v_2}, \dots, I_{v_k})}$$

$$\approx \mathcal{P}(I_{v_1} | w_i) \mathcal{P}(I_{v_2} | w_i) \dots \mathcal{P}(I_{v_k} | w_i) \mathcal{P}(w_i) \quad (3)$$

In equation (3), the conditional independence of the terms of form $\mathcal{P}(I_{v_k} | w_i)$ follows from the fact that, for a fixed world hypothesis w_i the camera views depend only on w_i and are independent of each other. However, for choosing the next best view (equation (2)), we cannot evaluate $\mathcal{P}(I_{v_{k+1}} | w_i)$. This is because we have not yet seen $I_{v_{k+1}}$. For this case, we generate a utility value for viewpoint v_{k+1} based on the expected number of occluded and visible apples for world model w_i (described in details in section VI-A).

A. Next Best View Planning

For predicting the next viewpoint, we need to evaluate the utility of each of the possible candidate views. Let $V_c = \{v_{k+1}, v_{k+2}, \dots, v_{k+r}\}$ be the set of candidate viewpoints. For each of this view points v_{k+j} , $j \in [1, r]$, we generate an image of each of our world hypotheses w_i . Our physical world models consist of ellipsoids $w_i = \{e_1^{w_i}, e_2^{w_i}, e_3^{w_i}, \dots, e_n^{w_i}\}$. We represent each of them in a quadric format which allows us to compute their projection quickly. While projecting, we sort the quadric by their depth order and compute the overlap between the projected ellipses. Following this procedure, we determine the visibility of each ellipsoid in the resulting image. If more than 20% of a projected ellipse is not occluded by any other we mark it as visible. Performing this operation for each viewpoint v_{k+j} and world model w_i we can determine the visible and occluded fruits for each view point for each world model w_i . We build two tables, visModel, occModel, of dimension $m \times p$ where m is the total number of viewpoints and p is the total number of world models.

$$\text{visModel}(i, j) = \text{visible apples generated by } w_j \text{ from } v_i$$

$$\text{occModel}(i, j) = \text{occluded apples generated by } w_j \text{ from view point } v_i$$

We define the utility of a view in terms of these visibility and occlusion models by building a similar table which we

refer to as the utilityModel.

$$\text{utilityModel}(i, j) = \frac{\text{visModel}(i, j)}{\text{occModel}(i, j) + \text{visModel}(i, j)}$$

Following the computation of this table, we normalize the utilities across all the view points to obtain a probability distribution of the utility for each world model. The next best view is found from these computed utilities. We consider two methods for this step. First, we plan for a single view that minimizes the entropy of our hypotheses set W . Second, we plan for a sequence of views which must be taken to verify the all different configurations of visible and occluded apples from our hypotheses set. It should be noted that regardless of the single/ multiple step planning we make one move at time (for multiple step - the first view in the sequence), recompute the posterior probabilities of world hypotheses and plan for the next view using the single/multiple step method. We start with the single step planning.

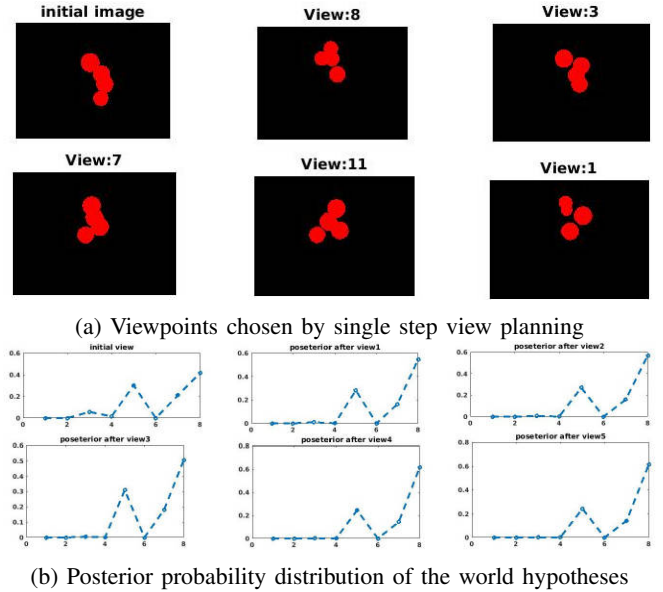


Fig. 5: Sample result from simulations. We randomly generated a 3D model of four apples. Our multiple step view planning algorithm selected three more views before converging to a world model with correct world counts

1) *Planning Single Step at a Time*: For single step planning we can directly use equation (2). For any candidate view point v_{k+1} , and the hallucinated image $I_{v_{k+1}}$ (generated by projecting world model w_i to the camera at view point v_{k+1}),

$$\mathcal{P}(w_i | I_{v_1}, I_{v_2}, \dots, I_{v_k}, I_{v_{k+1}}) \approx \mathcal{P}(I_{v_1} | w_i) \mathcal{P}(I_{v_2} | w_i) \dots \mathcal{P}(I_{v_k} | w_i) \mathcal{P}(I_{v_{k+1}} | w_i) \mathcal{P}(w_i)$$

$$= \mathcal{P}(w_i | I_{v_1}, I_{v_2}, \dots, I_{v_k}) \text{utilityModel}(k+1, i) \quad (4)$$

We replace this value in equation (2), after normalizing the values across all the view points.

2) *Planning Multiple Steps*: A simple way to do, multiple step planning is to plan a fixed number of views together. Essentially, it means considering $\binom{m}{k}$ candidates, where k is

the number of views to be planned together and m is the cardinality of the viewpoint sets. Consequently, it involves multiplying $k-1$ additional utility values to equation (4) and evaluating equation (2).

Instead of focusing on maximizing the number of visible apples, we can uncover the geometry of all possible world hypotheses from different view points. This is our second strategy for multiple step planning. For this, we find different configurations of visible and occluded apples from different viewpoints. Then we group the viewpoints for a particular world hypothesis w_i by the number of visible apples. Next, we find the maximum cardinality of these groups for all the world hypotheses and use this maximum cardinality g as the number of views to be planned together. We generate the candidate viewpoint sequence like the previous method and evaluate them in terms of the covered configurations of visible and occluded apples. The view sequences with the maximum score are evaluated using equation (2) and the best sequence is selected. The performance of the single and multiple step methods have been evaluated in terms of convergence and the average number of views required to converge. Figure 6 shows the results. We see that the multiple step method outperforms the single step method for cluster sizes 5 – 9. We explain the results further in section VII.

B. Updating the Posterior Probabilities of the World Model and Convergence Criterion

In this section we discuss, how we can incorporate information from new views to our world models. Essentially, with each additional view the posterior probabilities of our world models are updated. Suppose we have taken measurements from viewpoints $v_1, v_2, \dots, v_k$, and have chosen v_{k+1} as our next measurement location. Then the posterior for world model w_i is updated as follows

$$\mathcal{P}(w_i | I_{v_1}, I_{v_2}, \dots, I_{v_k} I_{v_{k+1}}) \approx \mathcal{P}(I_{v_1} | w_i) \mathcal{P}(I_{v_2} | w_i) \dots \mathcal{P}(I_{v_k} | w_i) \mathcal{P}(I_{v_{k+1}} | w_i) \mathcal{P}(w_i) \quad (5)$$

The exact value of the posterior is found by normalizing the likelihood across all the world models. Before we can apply equation (5) we need to resolve one more issue. Our sensing model does not directly provide the quantities of the form $\mathcal{P}(I_{v_k} | w_i)$. Instead, it provides the likelihood of a certain number of apples being visible in the input image. For the particular world model under consideration, we need to find the expected number of visible apples from the visibility model that was constructed in the previous section. After this step, we can associate the sensing likelihoods with $\mathcal{P}(I_{v_k} | w_i)$

We use the reduction in posterior entropy coupled with the difference between the successive peaks of the posterior as our termination criteria.

VII. EXPERIMENTAL RESULTS

To verify our algorithms, we conducted tests both in simulations and indoor environment. During the winter time

there are no apples on the trees and therefore field experiments were not possible. However, indoor experiments were arranged by using a realistic artificial apple tree (Figure 7a). For both simulation and indoor environments we assume that the fruit detection problem has been solved (i.e. we can locate fruit clusters in the input image). As currently there exists no method for actively planning views for counting apples, we compared our results against a lower bound provided by orthographic cameras. We also compared the performance of the single and multi-step view planning strategy.

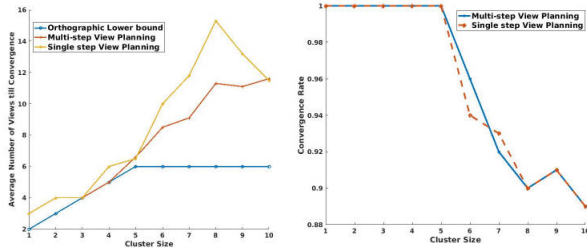
A. Comparison with Lower Bounds from an Orthographic Camera Using Cardinal Views

An orthographic camera can unfold the visible geometry of any object with six cardinal views: front, back, top, bottom, right and left. These views are often used in engineering drawings to capture the scale of objects. Essentially, an orthographic camera can compute the number of apples (visible from at least one view) in any cluster size with six views maximally. Given a fixed cluster size, we use the number of views required by an orthographic camera to be the lower bound for the number of views. For single apples, two perpendicular views are necessary to make sure that there is only a single apple. For two apples, we need three mutually perpendicular views, for three and four apples we need four and five views. For five or more apples six cardinal views are required. For orthographic cameras this bound is tight.

B. Simulation Results

We developed a simulation system using MATLAB for testing purposes. We use sets of randomly generated ellipsoids as the apples and simulate a camera with 40° field of view for capturing the images. To simulate the estimated depth from camera to the apple cluster we added Gaussian noise to the actual distance ($\sigma = 1, \mu = 0$). Using the estimated distance and estimated cluster center from image based measurements we define a viewing sphere (Figure. 2). Then 25 viewpoints are placed on the frontal hemisphere facing the camera. Using our world model generation procedure described in section V, we generate a set of world hypotheses W and predict the next best view points using single step and multiple step planning strategies. Here, we present simulation results for cluster size ranging from $[1, 10]$. We created 20 instances for each cluster size and computed the number of views required to converge for the single step and multiple step strategies. Figure 6 shows the comparison of the single step and multiple step strategy along with the orthographic lower bound and convergence plot. We see that the multiple step planner outperforms the single step planner for cluster sizes 5 – 9 (Fig. 6(a)). We see that for the large cluster sizes both planners have a higher chance of converging to wrong cluster size. This is attributed to our sensing model. In our previous work [3] we showed that, the accuracy of the likelihoods provided by our sensing model drops with increasing cluster size. However, in practice large clusters

are not very common and therefore handling them with 90% convergence is quite acceptable.



(a) Single step and multi-step view planning with orthographic lower bound (b) Convergence to the correct world model

Fig. 6: Simulation Results

C. Indoor Experiments

For the indoor experiments we use an artificial apple tree and synthetic apples. Figure 7 shows our experimental setup and how our algorithm performs in one of the experiments. The cluster under observation (Figure 7(b)) has three apples, but none of the views shows all of them. Our multi step planning algorithm plans the views required and converges to the correct world model containing three apples in four views.

VIII. CONCLUSION

In this paper, we considered a view planning task where the goal is to choose views so as to accurately count the number of fruits in a cluster. We presented an approach where the fruit are represented as ellipsoids. The number of ellipsoids, as well as their parameters (location and dimensions) constitute a world hypothesis. Rather than enumerating all possible world hypotheses, which is computationally prohibitive, we show how combinatorially distinct world hypotheses (regarding the visibility relationship between the apples) can be enumerated. Additionally, we present a likelihood function which associates a probability with each

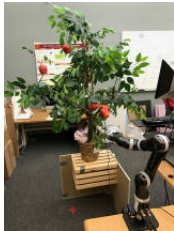
world hypothesis for a given view and propose greedy and multi-step planners which choose viewpoints to minimize the entropy of the posterior. These planners are implemented and evaluated both in simulation and in an indoor experimental setup. Our results indicate that for larger cluster size 5 – 9 the multi-step planner performs better than the single step planner. The results from simulations and indoor experiments are similar. It can be attributed to the fact that most of the difficulties in real world scenes are in sensing from individual images which is taken care of by our sensing model. In our future work, we plan to evaluate the system in an orchard during apple season (July – September in Minnesota) and also plan for the motion of the robot base.

IX. ACKNOWLEDGEMENT

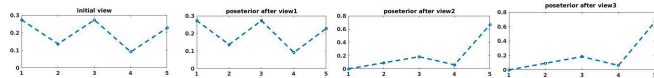
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REFERENCES

- [1] Q. Wang, S. Nuske, M. Bergerman, and S. Singh, "Automated crop yield estimation for apple orchards," in *Experimental Robotics*, ser. Springer Tracts in Advanced Robotics, J. P. Desai, G. Dudek, O. Khatib, and V. Kumar, Eds. Springer International Publishing, 2013, vol. 88, pp. 745–758.
- [2] S. Bargoti and J. P. Underwood, "Image segmentation for fruit detection and yield estimation in apple orchards," *CoRR*, vol. abs/1610.08120, 2016. [Online]. Available: <http://arxiv.org/abs/1610.08120>
- [3] P. Roy and V. Isler, "Vision-based apple counting and yield estimation," in *Experimental Robotics*. Springer, 2016.
- [4] —, "Surveying apple orchards with a monocular vision system," in *Automation Science and Engineering (CASE), 2016 IEEE International Conference on*. IEEE, 2016, pp. 916–921.
- [5] R. Bajcsy, "Active perception," *Proceedings of the IEEE*, vol. 76, no. 8, pp. 966–1005, 1988.
- [6] W. R. Scott, G. Roth, and J.-F. Rivest, "View planning for automated three-dimensional object reconstruction and inspection," *ACM Computing Surveys (CSUR)*, vol. 35, no. 1, pp. 64–96, 2003.
- [7] R. Bajcsy, Y. Aloimonos, and J. K. Tsotsos, "Revisiting active perception," *arXiv preprint arXiv:1603.02729*, 2016.
- [8] N. Atanasov, B. Sankaran, J. Le Ny, G. J. Pappas, and K. Daniilidis, "Nonmyopic view planning for active object classification and pose estimation," *IEEE Transactions on Robotics*, vol. 30, no. 5, pp. 1078–1090, 2014.
- [9] C. Potthast and G. S. Sukhatme, "A probabilistic framework for next best view estimation in a cluttered environment," *Journal of Visual Communication and Image Representation*, vol. 25, no. 1, pp. 148–164, 2014.
- [10] T. Patten, W. Martens, and R. Fitch, "Monte carlo planning for active object classification," *Autonomous Robots*, pp. 1–31, 2017.
- [11] J. Baur, C. Schütz, J. Pfaff, T. Buschmann, and H. Ulbrich, "Path planning for a fruit picking manipulator," in *Proceedings International Conference of Agricultural Engineering, Zürich, Switzerland*, 2014.
- [12] J. Underwood, M. Calleja, Z. Taylor, C. Hung, J. Nieto, R. Fitch, and S. Sukkarieh, "Real-time target detection and steerable spray for vegetable crops," in *proceedings, Workshop on Robotics in Agriculture at International Conference on Robotics and Automation (ICRA)*, 2015.



(a) Experimental Setup (b) Planned Views. None of them shows all three apples clearly.



(c) Posterior Probabilities of the World Model. The configuration of visible and hidden apples in the models $\{1, 2, 3, 4, 5\}$ are $\{(1, 0), (1, 1), (2, 0), (3, 0), (2, 1)\}$

Fig. 7: Indoor Experiments